

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Applied Business Analytics Practical
Practical 1: Introduction to Business Analytics:

AIM: a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.
b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

Function description

Function / Method	Description
<code>pd.read_csv()</code>	Loads CSV file into DataFrame.
<code>df.info()</code>	Shows dataset structure and data types.
<code>df.head()</code>	Displays first 5 rows.
<code>df.isnull().sum()</code>	Counts missing values.
<code>sns.countplot()</code>	Plots count of categorical data.
<code>plt.title()</code>	Adds title to plot.
<code>plt.show()</code>	Displays the graph.
<code>plt.figure()</code>	Creates figure with custom size.
<code>df.groupby()</code>	Groups data by a column.

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<code>.sum()</code>	Calculates total values.
<code>plot(kind='bar')</code>	Creates bar chart.
<code>pd.to_datetime()</code>	Converts to date format.
<code>.dt.month_name()</code>	Extracts month name.
<code>sns.histplot()</code>	Shows data distribution.
<code>df.corr()</code>	Calculates correlation matrix.
<code>sns.heatmap()</code>	Visualizes correlation matrix.

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('/content/supermarket_sales.csv')

print(df.info())
print(df.head())
```

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```
print(df.isnull().sum())

sns.countplot(x='Gender', data=df)
plt.title("Customer Gender Distribution")
plt.show()

sns.countplot(x='Customer type', data=df)
plt.title("Customer Type")
plt.show()

sns.countplot(x='Payment', data=df)
plt.title("Payment Method")
plt.show()

plt.figure(figsize=(10,5))
sns.countplot(y='Product line', data=df)
plt.title("Products Sold")
plt.show()

city_sales = df.groupby('City')['Total'].sum()
city_sales.plot(kind='bar')
plt.ylabel("Sales")
plt.title("Sales by City")
plt.show()

df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.month_name()

monthly = df.groupby('Month')['Total'].sum()
monthly.plot(kind='bar')
plt.title("Monthly Sales")
plt.ylabel("Sales")
plt.show()
```

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```
sns.histplot(df['Rating'], bins=10)
plt.title("Customer Ratings")
plt.show()

plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

Output:

```
*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Invoice ID             1000 non-null  object  
1   Branch                1000 non-null  object  
2   City                  1000 non-null  object  
3   Customer type         1000 non-null  object  
4   Gender                1000 non-null  object  
5   Product line          1000 non-null  object  
6   Unit price            1000 non-null  float64 
7   Quantity              1000 non-null  int64   
8   Tax 5%                1000 non-null  float64 
9   Total                 1000 non-null  float64 
10  Date                  1000 non-null  object  
11  Time                  1000 non-null  object  
12  Payment               1000 non-null  object  
13  cogs                  1000 non-null  float64 
14  gross margin percentage 1000 non-null  float64 
15  gross income          1000 non-null  float64 
16  Rating                1000 non-null  float64 
dtypes: float64(7), int64(1), object(9)
```

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```
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
None
```

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	

	Product line	Unit price	Quantity	Tax 5%	Total	Date	\
0	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	
1	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	
2	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	
3	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	
4	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	

```
***
```

	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	13:08	Ewallet	522.83	4.761905	26.1415	9.1
1	10:29	Cash	76.40	4.761905	3.8200	9.6
2	13:23	Credit card	324.31	4.761905	16.2155	7.4
3	20:33	Ewallet	465.76	4.761905	23.2880	8.4
4	10:37	Ewallet	604.17	4.761905	30.2085	5.3

Invoice ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Total	0
Date	0
Time	0
Payment	0
cogs	0
gross margin percentage	0
gross income	0
Rating	0

```
dtype: int64
```

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```
monthly = df.groupby("Month")["Total"].sum()
monthly.plot(kind='bar')
plt.title("Monthly Sales")
plt.xlabel("Sales")
plt.show()

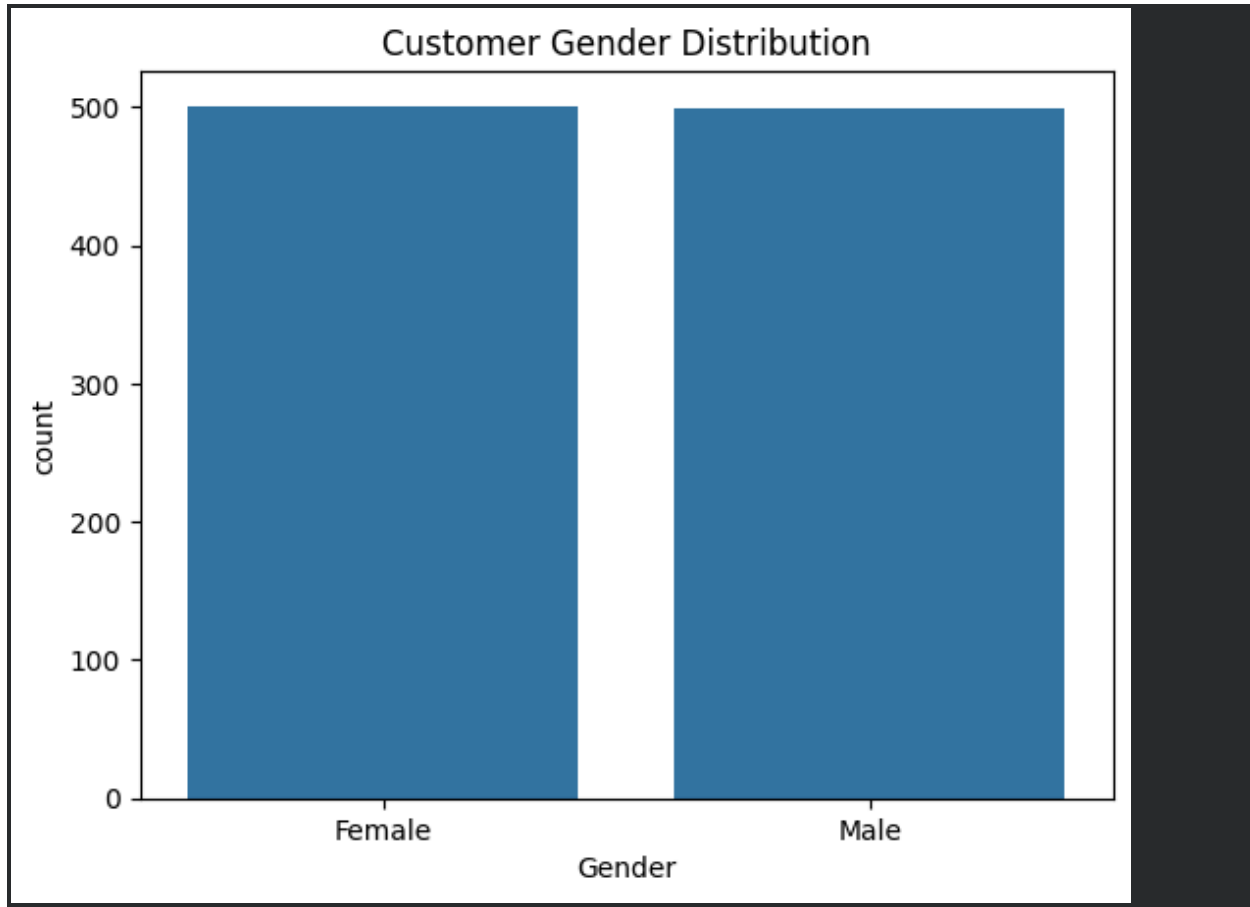
sns.histplot(df["Rating"], bins=10)
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sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	13:08	Buillet	522.83	4.761905	26.1415	9.1
1	18:29	Cash	76.48	4.761905	3.8200	9.6
2	13:23	Credit card	324.31	4.761905	16.2155	7.4
3	28:33	Buillet	465.76	4.761905	23.2880	8.4
4	18:37	Buillet	684.17	4.761905	38.2085	5.3

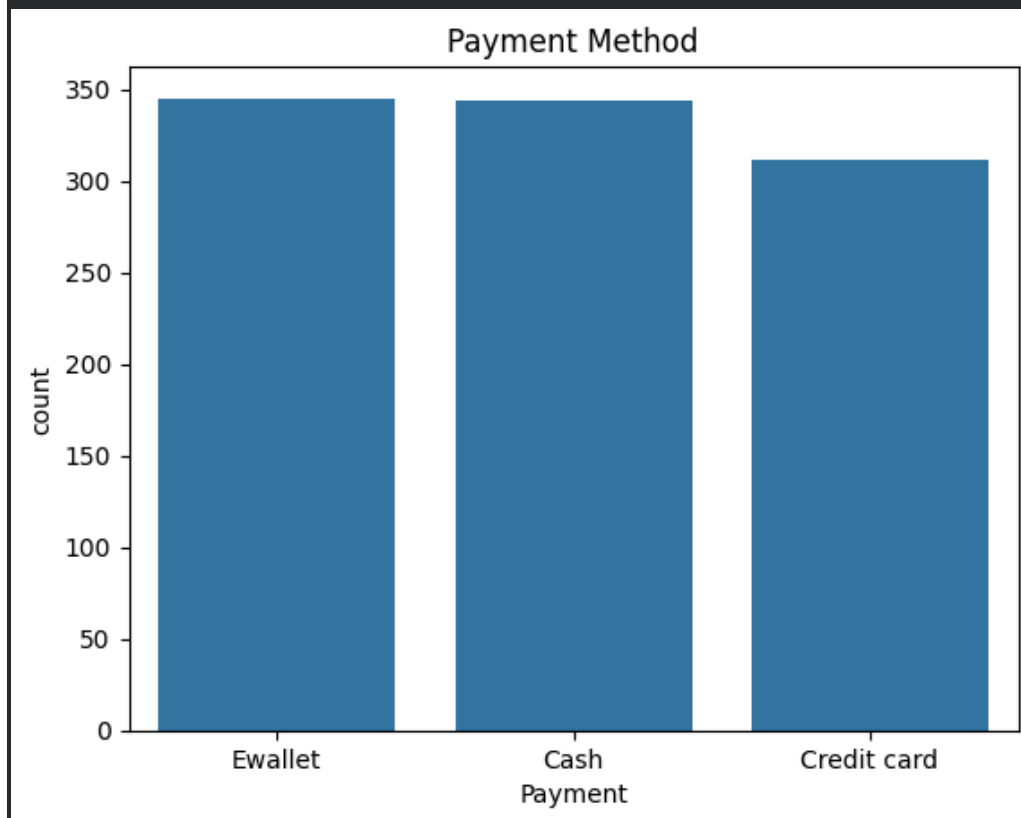
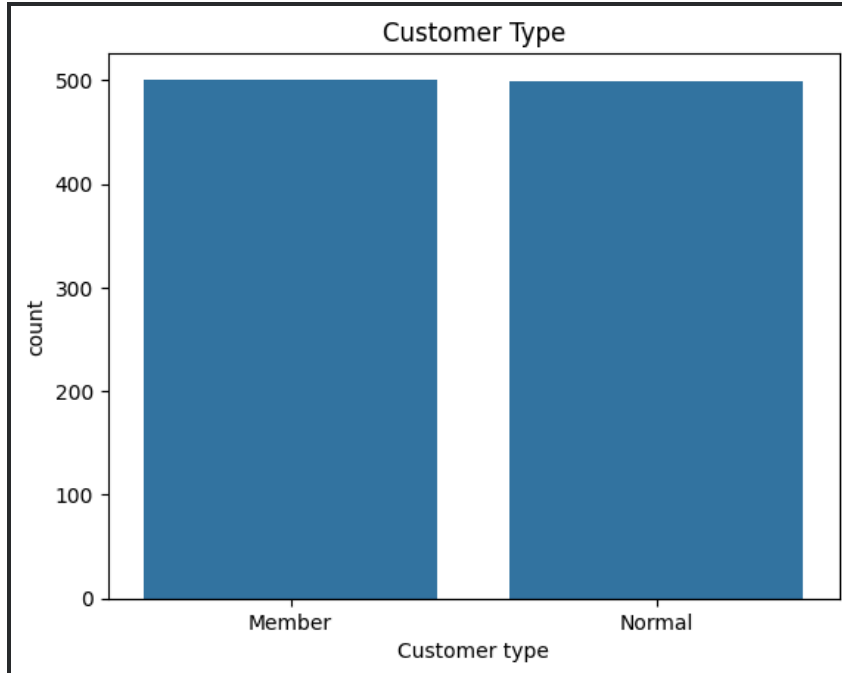
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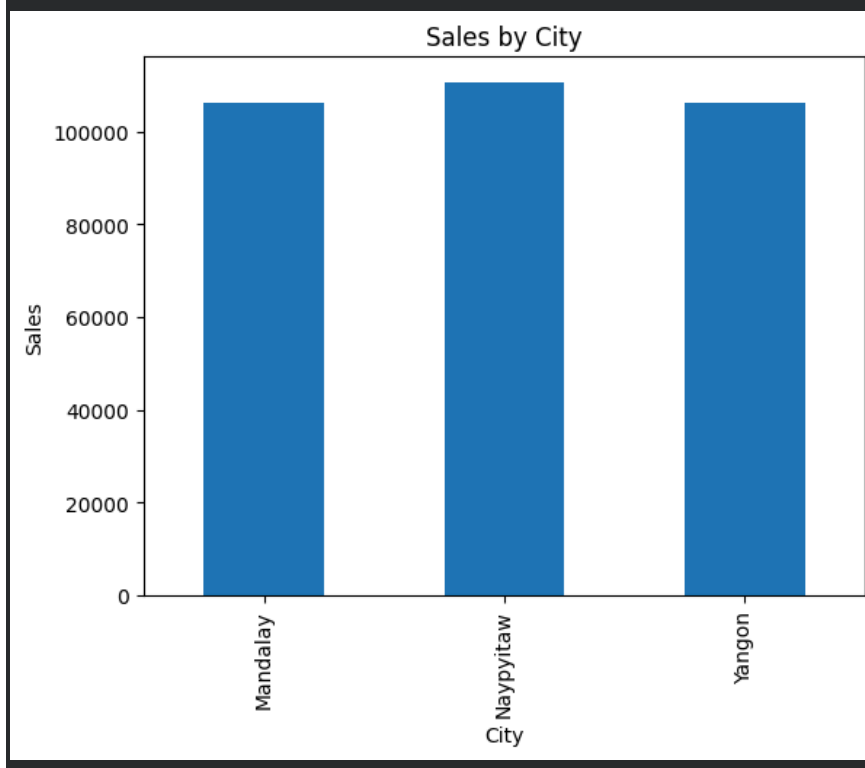
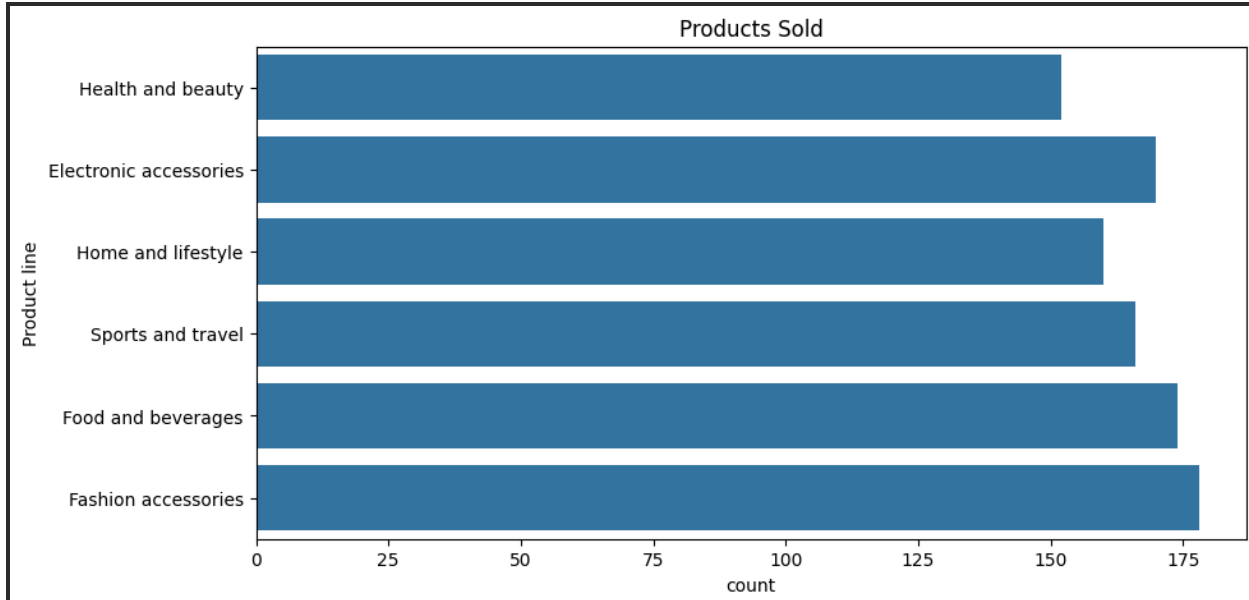
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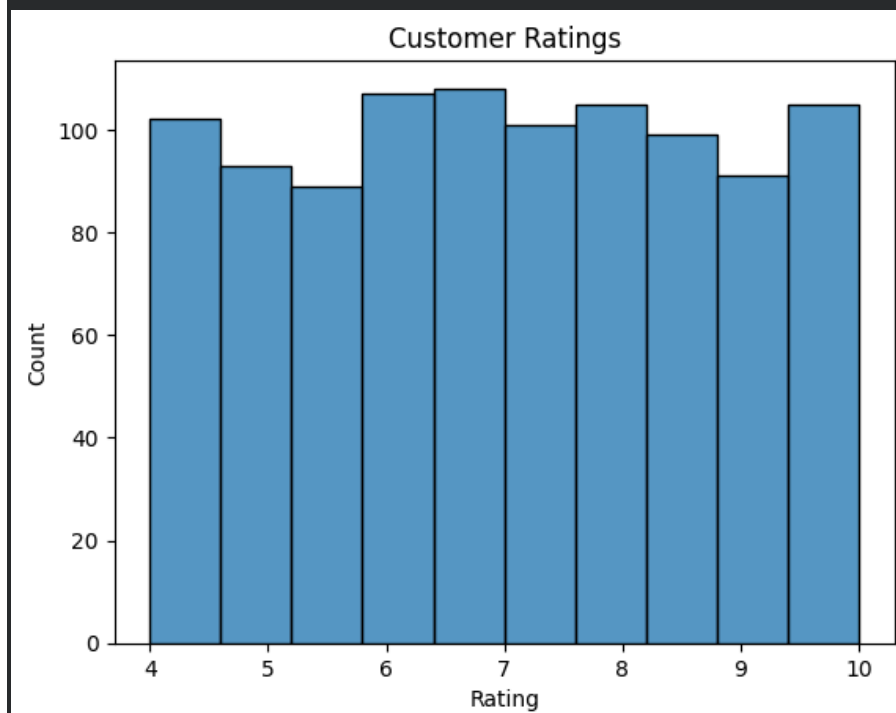
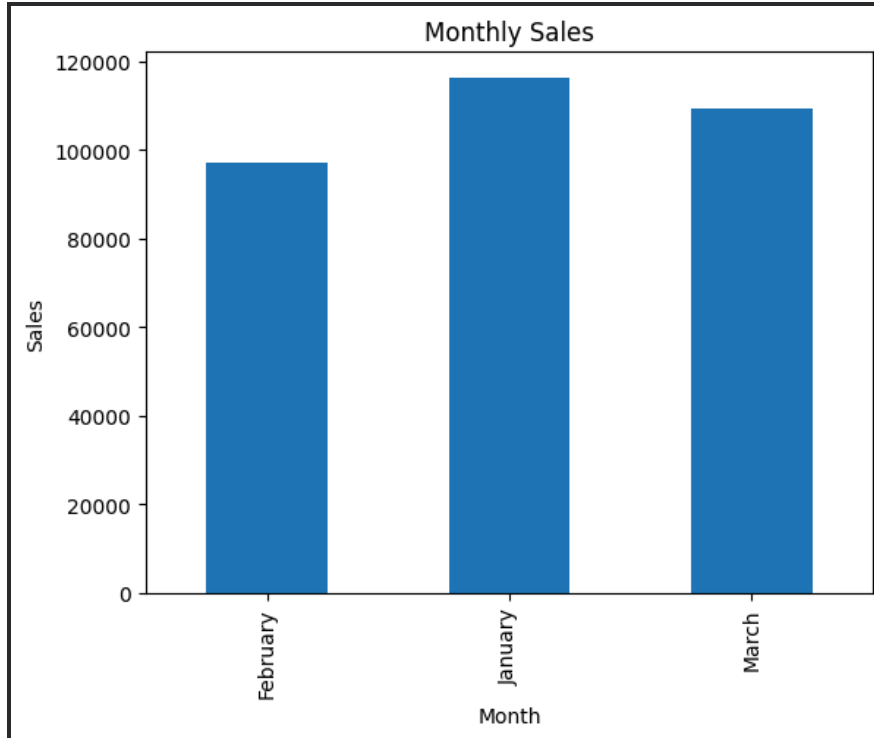
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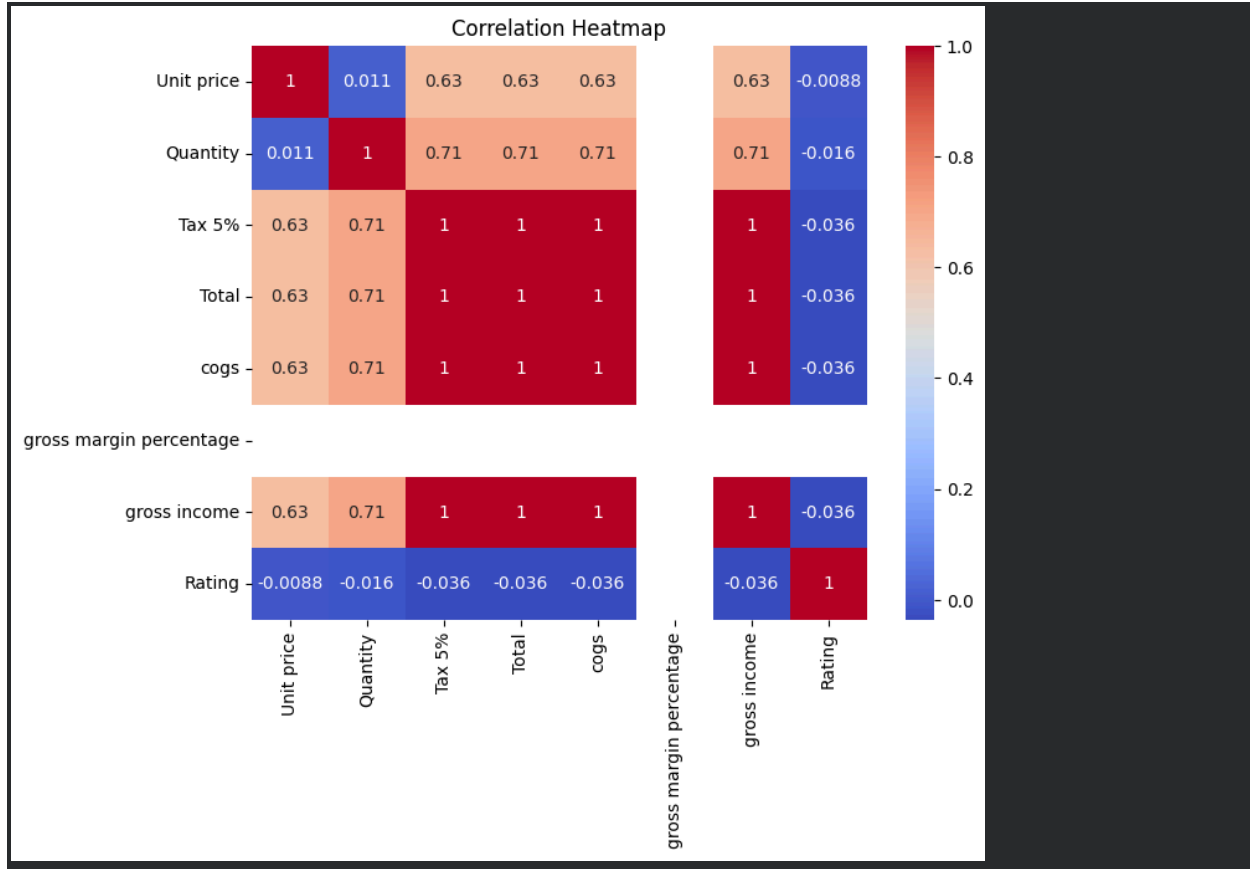
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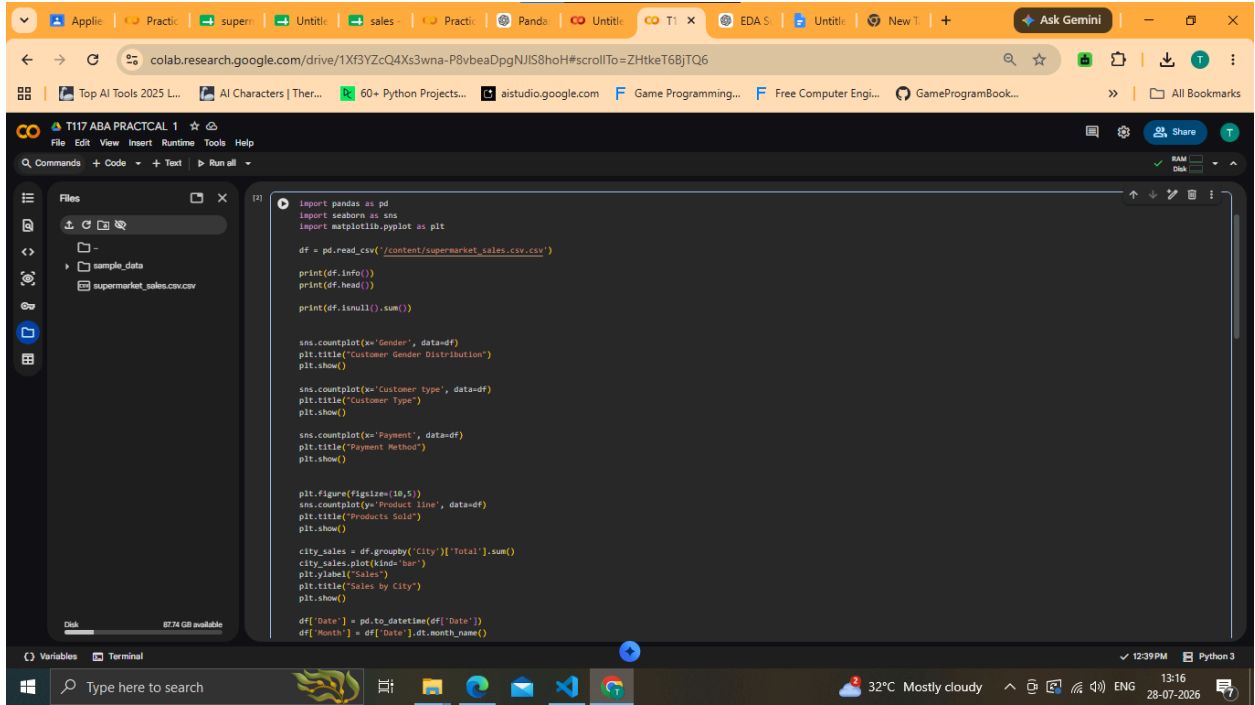
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The screenshot shows a Google Colab notebook interface. The browser address bar indicates the notebook is located at `colab.research.google.com/drive/1Xf3YZcQ4Xs3wna-P8vbeaDpgNjIS8hoH#scrollTo=ZHtkeT6BjTQ6`. The notebook title is "T117 ABA PRACTICAL 1". The left sidebar shows a file explorer with a folder named "sample_data" containing a file "supermarket_sales.csv". The main code editor contains the following Python code:

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```

The bottom status bar shows the system time as 12:39 PM, the temperature as 32°C, and the date as 28-07-2026. A large, light gray watermark "T117 ADARSH" is visible diagonally across the lower half of the image.

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